Technology Sector Equity Analysis: Pattern Discovery and Trading Strategy Development

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Objective:

The objective of this report is to analyze historical equity data from the technology sector, identify meaningful patterns and anomalies, propose a trading idea, and discuss methods for testing and evaluating the idea.

Sector Selected:

Technology Sector

Equities Selected:

- Apple (AAPL)
- Microsoft (MSFT)
- NVIDIA (NVDA)

Data Source:

Yahoo Finance historical stock data

Methodology:

Data Collection:

Historical daily stock prices were collected using Python and Yahoo Finance APIs.

Data Fields Used

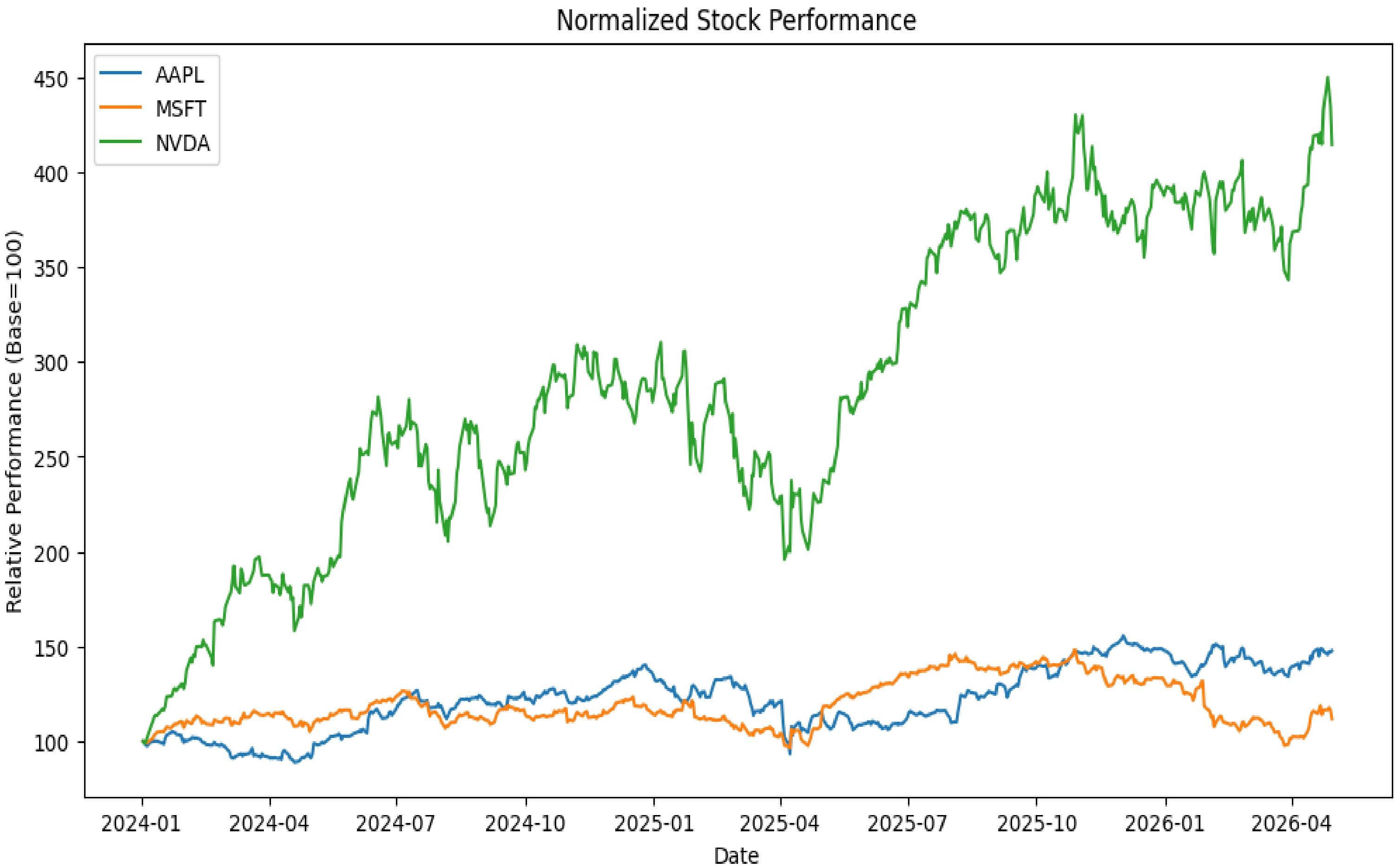
- Closing Price
- Daily Returns
- Trading Volume

Analysis Performed

- Price trend analysis
- Correlation analysis
- Return distribution analysis
- Moving average analysis

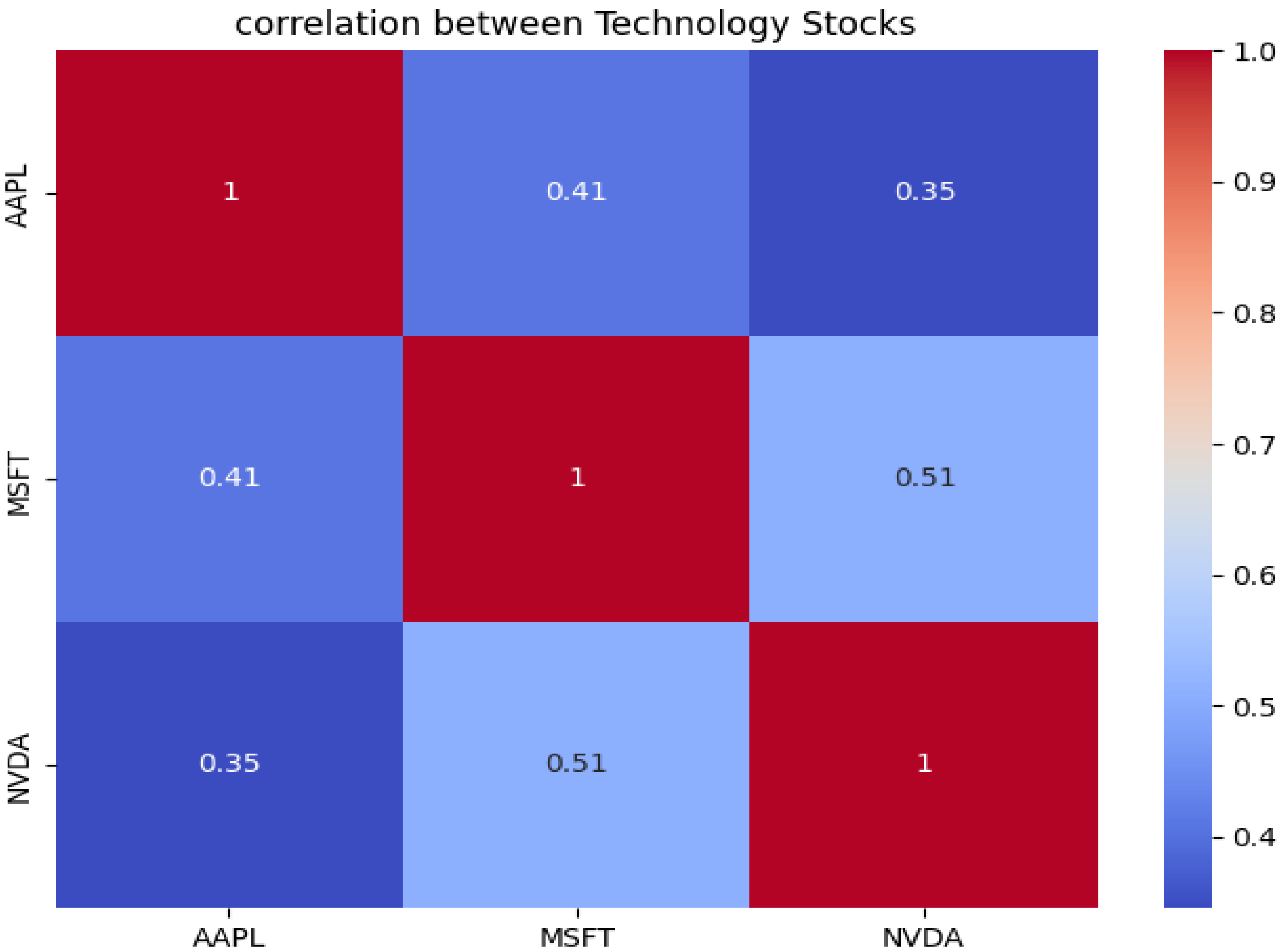
Pattern Analysis:

Pattern 1: NVIDIA significantly outperformed other technology stocks



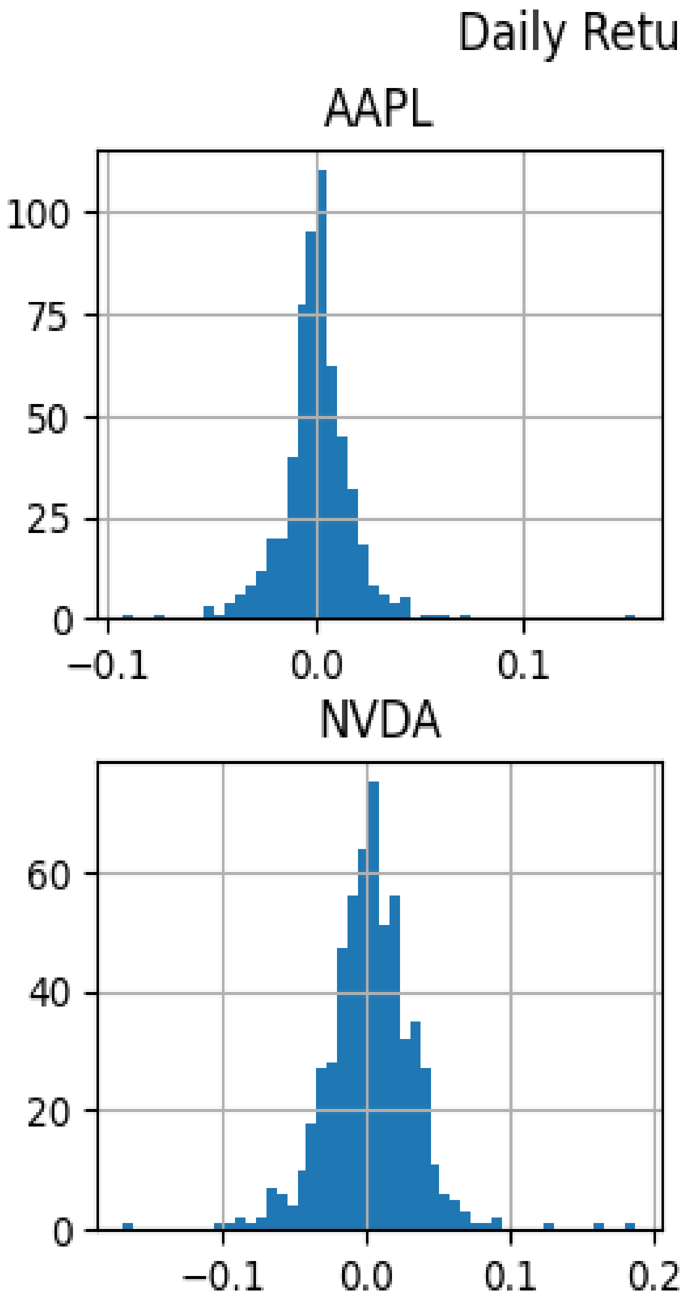
After normalizing all stocks to a common starting value, NVIDIA significantly outperformed Apple and Microsoft. NVIDIA reached approximately 450 compared with approximately 150 for the others. However, larger price fluctuations suggest greater risk and volatility.

Pattern 2: Moderate positive correlation among technology stocks



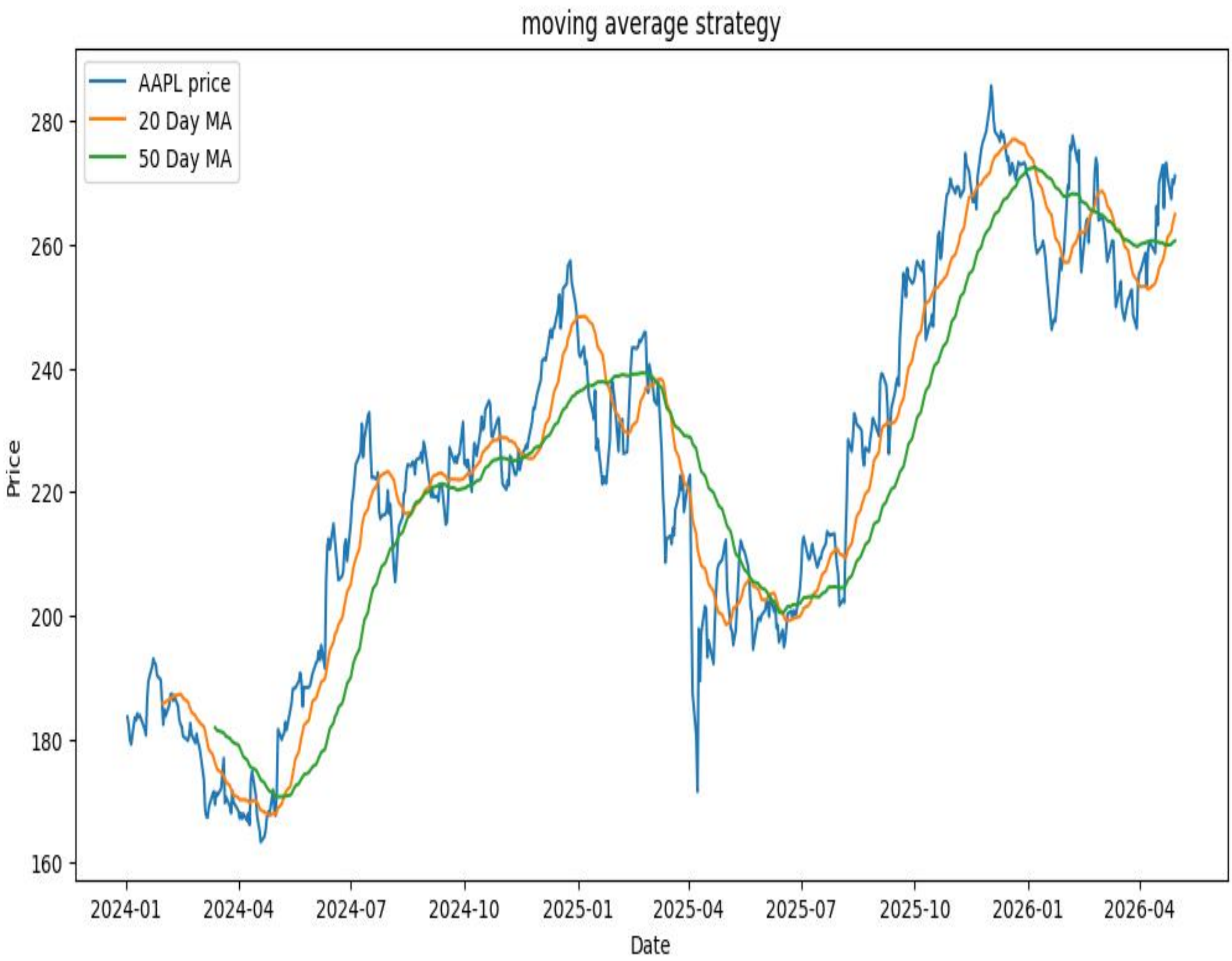
Microsoft–NVIDIA showed the highest correlation (~0.51), followed by Apple–Microsoft (~0.41). Technology stocks generally move together because of common market influences, although diversification benefits still exist.

Pattern 3: Higher volatility in NVIDIA



NVIDIA exhibited a wider spread of daily returns compared with Apple and Microsoft, indicating higher volatility.

Trading Idea: Moving Average Strategy



Buy signal

- 20-day moving average crosses above 50-day moving average

Sell signal

- 20-day moving average crosses below 50-day moving average

Reasoning: The strategy assumes that short-term price trends react faster than long-term trends. When the 20-day moving average crosses above the 50-day moving average, it may indicate strengthening market momentum and a potential upward trend.

Backtesting Approach

1. Collect historical stock data
2. Calculate moving averages
3. Generate trading signals
4. Compare against Buy-and-Hold strategy

Metrics:

- Total Return
- Sharpe Ratio
- Maximum Drawdown
- Win Rate

Evaluation Metrics:

- **Total Return:** Measures overall profitability.
- **Sharpe Ratio:** Measures return relative to risk taken.
- **Maximum Drawdown:** Measures largest decline from peak value.
- **Win Rate:** Percentage of profitable trades.

Backtest Results — MA Crossover Strategy (20/50 day)

Ticker	Strategy Return	Buy & Hold Return	Sharpe Ratio	Max Drawdown	Win Rate
AAPL	20.4%	47.7%	0.55	-24.6%	55.2%
MSFT	-9.1%	11.8%	-0.28	-19.4%	50.8%
NVDA	64.6%	314.6%	0.83	-23.5%	52.6%

The MA crossover strategy underperformed buy & hold on all three stocks. On NVDA, the strategy returned only 64.6% compared to 314.6% for buy & hold — because the signal is too slow for high-momentum stocks, entering after most of the move has already happened. On MSFT, the strategy produced a negative return of -9.1%, suggesting false signals in choppy, sideways markets. This indicates the strategy works best on stocks with slow, sustained trends.

Risks and Limitations

- False trading signals

- Overfitting
- Transaction costs
- External market events
- Limited sample size

AI Usage Statement

AI Usage Statement: AI was used to assist in structuring the analysis, improving explanations, and debugging code implementation. The stock selection, chart generation, interpretation of results, and validation of findings were performed by me.